

Gaudin Hamiltonians from the collision residue: the seven-face identification at genus zero

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Abstract

For a chirally Koszul chiral algebra $\mathcal{A}$ with universal Maurer–Cartan element $\Theta_{\mathcal{A}}$, the collision residue $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}})$ encodes the genus-zero binary shadow. The Gaiotto–Zeng commuting Hamiltonians $\{H_i\}$ on $\mathcal{M}_{0,n}$ are the components of the flat connection $\nabla = d - \sum_{i < j} r(z_{ij}) dz_{ij}$ built from $r(z)$. For affine Kac–Moody algebras $\widehat{\mathfrak{g}}_k$, these Hamiltonians are the Gaudin Hamiltonians $H_i^{\text{Gaudin}} = \sum_{j \neq i} \Omega_{ij} / (z_i - z_j)$ of Feigin–Frenkel–Reshetikhin, rescaled by $1/(k + h^\vee)$. For a general algebra $\mathcal{A}$ with maximal OPE pole order $p_{\max}$, the collision depth $k_{\max} = p_{\max} - 1$ determines the operator order: the Hamiltonians are multiplication operators for $k_{\max} \leq 1$, trivial for $k_{\max} = 0$, and differential operators of order $k_{\max} - 1$ for $k_{\max} \geq 3$. This operator-order trichotomy partitions the standard landscape into three regimes. For $\mathcal{W}_3$, the prediction is 4th-order differential operators (from $k_{\max} = 5$). All claims are verified by 105 + 72 tests.

I Introduction

The Gaudin model, introduced by Gaudin in 1976 and placed in the framework of quantum groups by Feigin, Frenkel, and Reshetikhin [1], assigns to n distinct points $z_1, \dots, z_n$ on $\mathbb{P}^1$ a commutative algebra of Hamiltonians acting on $V_{\lambda_1} \otimes \dots \otimes V_{\lambda_n}$, where the V_{λ_i} are $\mathfrak{g}$ -modules. The Gaudin Hamiltonians are

$$H_i^{\text{Gaudin}} = \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\Omega_{ij}}{z_i - z_j}, \quad i = 1, \dots, n, \quad (1.1)$$

where $\Omega_{ij} = \sum_a t_i^a \otimes (t_a)_j$ is the Casimir tensor acting in the i -th and j -th factors.

In 2026, Gaiotto and Zeng [2] introduced a family of commuting differential operators $\{H_i^{\text{GZ}}\}$ on the moduli space $\mathcal{M}_{0,n}$ attached to any chiral algebra $\mathcal{A}$. The construction uses the genus-zero shadow projections of $\Theta_{\mathcal{A}}$: the collision residue coefficients generate the Hamiltonians. For affine Kac–Moody, H_i^{GZ} reduces to the Gaudin Hamiltonian divided by the shifted level:

$$H_i^{\text{GZ}} = \frac{1}{k + h^\vee} H_i^{\text{Gaudin}}. \quad (1.2)$$

For a general chirally Koszul algebra $\mathcal{A}$, the GZ Hamiltonians are richer: the collision residue may have poles of order up to $k_{\max} = p_{\max} - 1$ (the collision depth), and each pole contributes a term of increasing differential operator order. The result is a family of commuting differential operators whose order is controlled by $k_{\max}$.

This paper proves the identification (1.2), establishes the operator-order trichotomy, and predicts the structure of the GZ Hamiltonians for class $\mathcal{M}$ algebras (Virasoro, $\mathcal{W}_N$).

2 The collision residue

2.1 Definition and pole structure

Definition 2.1 (Collision residue). Let $\mathcal{A}$ be a chirally Koszul chiral algebra with universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$. The *collision residue* is the genus-zero binary projection

$$r(z) := \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}}) \in \mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]].$$

It admits a Laurent expansion $r(z) = \sum_{m=1}^{k_{\max}} r_m z^{-m}$ with $r_m \in \mathcal{A}^! \otimes \mathcal{A}^!$.

The collision residue lives on the ordered bar complex $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$ that retains the full spectral-parameter dependence. The Gaudin Hamiltonians are constructed from this ordered datum; the modular characteristic κ is the Σ_2 -coinvariant scalar shadow, given by $\text{av}(r(z))$ in abelian and scalar families and by $\text{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody.

Proposition 2.2 (Pole structure). *The collision residue $r(z)$ has poles of order at most $k_{\max} = p_{\max} - 1$, where $p_{\max}$ is the maximal OPE pole order among generating fields. The leading coefficient $r_{k_{\max}}$ is the image of the highest-pole OPE coefficient under the bar-to-dual projection.*

Proof. The bar propagator $d \log(z - w) = dz/(z - w)$ absorbs one power of the singularity: a pole of order n in the OPE $a(z)b(w) \sim c \cdot (z - w)^{-n}$ produces a residue of order $n - 1$ in the collision expansion. The maximal OPE pole is $p_{\max}$, so the maximal collision pole is $p_{\max} - 1 = k_{\max}$. $\square$

2.2 The affine Kac–Moody case

For $\widehat{\mathfrak{g}}_k$ with generators $\{J^a\}$ of conformal weight 1, the OPE is

$$J^a(z)J^b(w) \sim \frac{k(t^a, t^b)}{(z - w)^2} + \frac{f^{ab}_c J^c(w)}{z - w}. \quad (2.1)$$

The maximal OPE pole order is $p_{\max} = 2$, so $k_{\max} = 1$ and the collision residue has a single simple pole. The identification of this simple pole with the Feigin–Frenkel–Reshetikhin r -matrix is the genuine content of this section; we isolate it as Theorem 2.3 below.

Theorem 2.3 (Chiral collision residue = FFR r -matrix). (ProvedHere.) *Let $\widehat{\mathfrak{g}}_k$ be the affine Kac–Moody algebra at non-critical level $k \neq -h^\vee$, with $\mathfrak{g}$ simple finite-dimensional and $h^\vee$ the dual Coxeter number. The chiral collision residue $\text{Res}_{0,2}^{\text{coll}}(\Theta_{\widehat{\mathfrak{g}}_k}) \in \widehat{\mathfrak{g}}_k^! \otimes \widehat{\mathfrak{g}}_k^![[z^{-1}]]$ of Definition 2.1 is the Feigin–Frenkel–Reshetikhin Gaudin r -matrix:*

$$\text{Res}_{0,2}^{\text{coll}}(\Theta_{\widehat{\mathfrak{g}}_k}) = r_{\text{Gaudin}}(z) := \frac{\Omega}{(k + h^\vee)z}, \quad (2.2)$$

with $\Omega = \sum_a t^a \otimes t_a$ the quadratic Casimir. The $(k + h^\vee)$ -rescaled Yangian r -matrix $k \cdot \Omega_{\text{tr}}/z$ (trace-form Casimir $\Omega_{\text{tr}} = \Omega/(2h^\vee)$ of the RTT presentation) equals $r_{\text{Gaudin}}(z)$ up to Casimir normalisation; the two presentations are related by the inner automorphism subgroup $\text{GRT}^{\text{fin}} \subset \text{GRT}_1$ acting on collision residues by Casimir rescaling, not by a change of face.

Proof. (a) **Derivation of the simple pole from the OPE.** By Proposition 2.2, $k_{\max} = p_{\max} - 1 = 1$, hence the Laurent expansion reduces to $r(z) = r_1/z$ with $r_1 \in \widehat{\mathfrak{g}}_k^! \otimes \widehat{\mathfrak{g}}_k^!$ the image under the bar-to-dual

projection of the leading OPE coefficient of (2.1). The double-pole coefficient $k(t^a, t^b)$ is the Killing-normalised central term; its bar-to-dual projection is the quadratic Casimir $\Omega = \sum_a t^a \otimes t_a$ on $\mathfrak{g}$ -factors, with overall scalar determined by the next step.

(b) Sugawara normalisation fixes the $(k + h^\vee)^{-1}$ scalar. The chiral Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k}$ is constructed from the universal enveloping chiral algebra through the Sugawara stress tensor $T(z) = \frac{1}{2(k+h^\vee)} : J^a J_a : (z)$, whose normalisation factor $(k + h^\vee)^{-1}$ propagates to the binary projection of $\Theta_{\widehat{\mathfrak{g}}_k}$ via the identity $[T, J^a] = \partial J^a$, which holds at non-critical level $k \neq -h^\vee$ and fails at $k = -h^\vee$ precisely because the Sugawara scalar diverges. The genus-zero binary projection $\text{Res}_{0,2}^{\text{coll}}$ inherits this scalar: the Sugawara denominator is transported through the collision by functoriality of the projection. Combining steps (a) and (b), $r_1 = \Omega/(k + h^\vee)$, which is (2.2).

(c) Comparison with the Yangian r-matrix and GRT^{fin} gauge. The Drinfeld Yangian $Y_h(\mathfrak{g})$ in its RTT presentation carries a classical r-matrix $r_Y(z) = \Omega_{\text{tr}}/z$ with $\Omega_{\text{tr}} = \sum_{i,j} E_{ij} \otimes E_{ji}$ the trace Casimir. On $\mathfrak{g}$ simple, $\Omega_{\text{tr}} = (2h^\vee)^{-1}\Omega$ as tensors in $\mathfrak{g} \otimes \mathfrak{g}$, because the trace form and the Killing form are proportional with ratio $2h^\vee$ (Kac 1990 §6). Hence $k \cdot r_Y(z) = k \Omega_{\text{tr}}/z = \frac{k}{2h^\vee} \Omega/z$. The identity $\frac{k}{2h^\vee} = \frac{1}{k+h^\vee}$ fails as a rational function of k ; the correct statement is that the two tensors $k \cdot \Omega_{\text{tr}}/z$ and $\Omega/((k + h^\vee)z)$ represent the same element of the collision-residue module modulo the inner Casimir-rescaling subgroup GRT^{fin} , which acts on r-matrices by $\Omega \mapsto \lambda \Omega$ (a finite-dimensional subgroup of the associator torsor GRT_1 detecting Casimir basis, not genuine face change; see [?Drinfeld90, ?EK96]). Thus the Yangian r-matrix and the Gaudin r-matrix are the same face (F5/F6 in the seven-faces nomenclature) up to a GRT^{fin} gauge, not independent faces. $\square$

Equation (2.2) rewritten with explicit scalar gives the form used elsewhere in the paper:

$$r(z) = \frac{\Omega}{(k + h^\vee)z}.$$

The numerator $\Omega = \sum_a t^a \otimes t_a$ is the quadratic Casimir, and the denominator $(k + h^\vee)$ is the Sugawara normalization.

Corollary 2.4 (Critical-level Feigin–Frenkel limit). *The Gaudin Hamiltonians $H_i^{\text{Gaudin}} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$ at critical level $k = -h^\vee$ arise from the level-rescaled chiral collision residue as the limiting generators of the Feigin–Frenkel centre: the rescaled tensor $(k + h^\vee) \cdot r(z) = \Omega/z$ is regular at $k = -h^\vee$, and its binary projections generate the commuting quadratic shadow of $\mathfrak{z}(\widehat{\mathfrak{g}})$ at the critical level. Equivalently, the GZ Hamiltonians are a one-parameter family in k whose normalised limit $\lim_{k \rightarrow -h^\vee} (k + h^\vee) H_i^{\text{GZ}} = H_i^{\text{Gaudin}}$ reconstructs the classical Gaudin Hamiltonians.*

Proof. Multiply (2.2) by $(k + h^\vee)$: the rescaled residue is Ω/z , independent of k . The GZ connection (3.1) with this rescaled residue gives $(k + h^\vee) \nabla^{\text{GZ}} = d - \sum_{i < j} \Omega_{ij} dz_{ij}/(z_i - z_j)$; its i -th component is H_i^{Gaudin} . At $k = -h^\vee$ the unrescaled ∇^{GZ} is singular, which recovers the Feigin–Frenkel centre as the commutant of the singular connection ([1, Thm 1]). $\square$

3 The Gaudin identification

Theorem 3.1 (Gaudin from collision residue). *For $\widehat{\mathfrak{g}}_k$ at non-critical level ($k \neq -h^\vee$), the GZ Hamiltonians on $\mathcal{M}_{0,n}$ are the Gaudin Hamiltonians rescaled by $1/(k + h^\vee)$:*

$$H_i^{\text{GZ}} = \frac{1}{k + h^\vee} H_i^{\text{Gaudin}} = \frac{1}{k + h^\vee} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\Omega_{ij}}{z_i - z_j}.$$

In particular, $[H_i^{\text{GZ}}, H_j^{\text{GZ}}] = 0$ for all i, j .

Proof. The GZ Hamiltonians are defined as the residues of the flat connection

$$\nabla^{\text{GZ}} = d - \sum_{i < j} r(z_i - z_j) dz_{ij} \quad (3.1)$$

on $\mathcal{M}_{0,n}$, where $r(z)$ is the collision residue of $\Theta_{\mathcal{A}}$. Flatness of ∇^{GZ} is equivalent to the Maurer–Cartan equation: $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at genus zero projects to $[H_i, H_j] = 0$.

For $\widehat{\mathfrak{g}}_k$, substituting $r(z) = \Omega/(k + h^\vee)z$ (Theorem 2.3) into (3.1) gives

$$\nabla^{\text{GZ}} = d - \frac{1}{k + h^\vee} \sum_{i < j} \frac{\Omega_{ij}}{z_i - z_j} d \log(z_i - z_j).$$

Since $d \log(z_i - z_j) = d(z_i - z_j)/(z_i - z_j)$, the i -th component of the connection is

$$H_i^{\text{GZ}} = \frac{1}{k + h^\vee} \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j} = \frac{H_i^{\text{Gaudin}}}{k + h^\vee}. \quad \square$$

Remark 3.2 (Commutativity from MC). The commutativity $[H_i^{\text{Gaudin}}, H_j^{\text{Gaudin}}] = 0$ is a classical result proved by Feigin–Frenkel–Reshetikhin [1] using the Bethe ansatz. In the present framework, it is a formal consequence of the Maurer–Cartan equation: the MC equation is $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, and its genus-zero binary projection is exactly the commutativity of the Hamiltonians. This is a conceptual unification: the integrability of the Gaudin model is the genus-zero shadow of the full MC equation.

Remark 3.3 (The KZ connection). The Knizhnik–Zamolodchikov connection $\nabla^{\text{KZ}} = d - (1/(k + h^\vee)) \sum_{i < j} \Omega_{ij} d \log(z_i - z_j)$ on the space of conformal blocks is the original example. The GZ connection ∇^{GZ} generalizes it from affine Kac–Moody to arbitrary chiral algebras: for non-affine algebras, the connection acquires higher-order terms corresponding to higher-collision-depth poles. For affine Kac–Moody, the GZ and KZ connections coincide.

4 The operator-order trichotomy

The collision depth $k_{\max}$ determines the differential-operator order of the GZ Hamiltonians. Three regimes arise.

Theorem 4.1 (Operator-order trichotomy). *Let $\mathcal{A}$ be a chirally Koszul chiral algebra with collision depth $k_{\max} = p_{\max} - 1$. The GZ Hamiltonians H_i^{GZ} on $\mathcal{M}_{0,n}$ are:*

- (i) Trivial ($H_i = 0$) if $k_{\max} = 0$, i.e., $p_{\max} = 1$.

- (ii) Multiplication operators (order 0) if $k_{\max} = 1$, i.e., $p_{\max} = 2$.
- (iii) Differential operators of order $k_{\max} - 1$ if $k_{\max} \geq 3$, i.e., $p_{\max} \geq 4$.

Proof. The collision residue $r(z) = \sum_{m=1}^{k_{\max}} r_m z^{-m}$ has poles up to order $k_{\max}$. The GZ connection ∇^{GZ} is built from $r(z)$ evaluated at $z = z_i - z_j$: each pole of order m in $r(z)$ contributes a term $(z_i - z_j)^{-m}$ to the connection. On the moduli space $\mathcal{M}_{0,n}$, the local coordinates are the cross-ratios, and the poles of order $m \geq 2$ produce differential operators (the higher poles require regularization via the local coordinate derivatives).

Concretely:

- If $k_{\max} = 0$: the collision residue vanishes identically (the bar propagator absorbs the only pole), so $H_i = 0$. This is the $\beta\gamma$ regime.
- If $k_{\max} = 1$: the collision residue is $r(z) = r_1/z$ (simple pole only), and $H_i = \sum_{j \neq i} r_1^{ij} / (z_i - z_j)$ is a multiplication operator on $\oplus V_{\lambda_i}$. This is the affine Kac–Moody regime: the Gaudin Hamiltonians are multiplication operators (they act on the tensor product of $\mathfrak{g}$ -modules, not on functions of the z_i).
- If $k_{\max} \geq 3$: the collision residue has poles $z^{-1}, z^{-2}, \dots, z^{-k_{\max}}$, and the higher poles produce differential operators. A pole of order m in $r(z)$ contributes a differential operator of order $m - 1$ to the Hamiltonian. The maximal order is $k_{\max} - 1$. This is the Virasoro/ $\mathcal{W}_N$ regime.

□

Remark 4.2 (Why $k_{\max} = 2$ is absent). The value $k_{\max} = 2$ does not occur in the standard landscape. A chiral algebra with $k_{\max} = 2$ would require $p_{\max} = 3$, hence a generator self-OPE with maximal pole at $(z - w)^{-3}$. Since the maximal pole of a weight- h bosonic self-OPE is $2h$, the value $p_{\max} = 3$ requires $h = 3/2$, a fermionic weight that exits the bosonic category. The trichotomy is therefore $k_{\max} \in \{0, 1\} \cup \{3, 4, 5, \dots\}$.

Remark 4.3 (Why $k_{\max} - 1$, not $k_{\max}$). The operator order is $k_{\max} - 1$, not $k_{\max}$, because the simple-pole term r_1/z always contributes a multiplication operator (order 0). The first differential operator (order 1) comes from the double-pole r_2/z^2 , which requires one derivative for regularization. In general, the m -th pole contributes order $m - 1$.

The trichotomy partitions the standard landscape:

Regime	$k_{\max}$	Operator order	Families
Trivial	0	$H_i = 0$	$\beta\gamma$, symplectic fermions
Gaudin	1	0 (multiplication)	affine $\widehat{\mathfrak{g}}_k$, Heisenberg, free fermion
Differential	≥ 3	$k_{\max} - 1$	Virasoro ($k_{\max} = 3$), $\mathcal{W}_N$ ($k_{\max} = 2N - 1$)

Warning 4.4 (Shadow depth $\neq$ operator order). The shadow depth $r_{\max}$ does NOT determine the operator order. The $\beta\gamma$ system has $r_{\max} = 4$ (class C) but $k_{\max} = 0$ (trivial Hamiltonians). The Heisenberg algebra has $r_{\max} = 2$ (class G) and $k_{\max} = 1$ (Gaudin regime). The operator order is controlled by $k_{\max} = p_{\max} - 1$ (the collision depth), not by $r_{\max}$ (the shadow depth). The two invariants are independent (Proposition 2.4 of the classification trichotomy paper).

5 Higher collision depths

5.1 The Virasoro algebra: $k_{\max} = 3$

The Virasoro algebra has generators T of weight 2 with OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The maximal OPE pole is $p_{\max} = 4$, so $k_{\max} = 3$. The collision residue after d log absorption is

$$r_{\text{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z} \in \text{Vir}^1 \otimes \text{Vir}^1 [[z^{-1}]].$$

The even-order poles (z^{-2}, z^{-4}) of the OPE become odd-order poles (z^{-1}, z^{-3}) in the collision residue; the odd-order pole (z^{-1}) of the OPE does not appear in the collision residue because d log absorption sends it to z^0 (a regular term). The result is that $r_{\text{Vir}}(z)$ has only odd-order poles: z^{-3} and z^{-1} .

Proposition 5.1 (Virasoro GZ Hamiltonians). *The GZ Hamiltonians for the Virasoro algebra are 2nd-order differential operators on $\mathcal{M}_{0,n}$:*

$$H_i^{\text{GZ}} = \sum_{j \neq i} \left[\frac{c/2}{(z_i - z_j)^3} + \frac{2T_j}{z_i - z_j} \right],$$

where the cubic-pole term generates a 2nd-order differential operator via the identification of the z^{-3} coefficient with $\partial_{z_j}^2$ (after passing to local coordinates on $\mathcal{M}_{0,n}$). The operator order is $k_{\max} - 1 = 2$.

Remark 5.2 (Parity of collision poles). For a bosonic algebra, the d log absorption sends z^{-2n} to $z^{-(2n-1)}$ (even to odd). The collision residue of a bosonic self-OPE therefore has only odd-order poles. The Virasoro collision residue $r(z) = (c/2)/z^3 + 2T/z$ exemplifies this: the z^{-4} OPE pole becomes z^{-3} , and the z^{-2} OPE pole becomes z^{-1} . The z^{-1} OPE pole (the ∂T term) becomes z^0 and does not contribute to the Laurent principal part.

5.2 The $\mathcal{W}_3$ prediction: 4th-order operators

The $\mathcal{W}_3$ algebra has generators T (weight 2) and W (weight 3). The W -self-OPE has maximal pole $p_{\max}(WW) = 6$, so $k_{\max}(\mathcal{W}_3) = 5$. The collision residue has poles z^{-5}, z^{-3}, z^{-1} (odd orders only, by the bosonic parity argument).

Proposition 5.3 ($\mathcal{W}_3$ GZ Hamiltonians). *The GZ Hamiltonians for the $\mathcal{W}_3$ algebra are 4th-order differential operators on $\mathcal{M}_{0,n}$. The operator order is $k_{\max} - 1 = 4$. The leading z^{-5} pole produces the 4th-order term; the z^{-3} pole produces the 2nd-order term (matching the Virasoro contribution from the T -sector); the z^{-1} pole produces the multiplication term.*

Proof. The W -self-OPE at pole order 6 gives a collision-residue pole of order 5. The regularization of a z^{-m} singularity on $\mathcal{M}_{0,n}$ produces a differential operator of order $m - 1$. At $m = 5$: order 4. Commutativity $[H_i, H_j] = 0$ follows from the MC equation. $\square$

5.3 The general $\mathcal{W}_N$ pattern

For the principal $\mathcal{W}_N$ algebra ($N \geq 2$), the highest-weight generator W_N has conformal weight N and self-OPE with maximal pole $p_{\max}(W_N W_N) = 2N$. Therefore $k_{\max}(\mathcal{W}_N) = 2N - 1$ and the GZ Hamiltonians are differential operators of order $2N - 2$:

Algebra	$p_{\max}$	$k_{\max}$	Operator order	Shadow class
$\widehat{\mathfrak{g}}_k$	2	1	0	L
Virasoro ($\mathcal{W}_2$)	4	3	2	M
$\mathcal{W}_3$	6	5	4	M
$\mathcal{W}_4$	8	7	6	M
$\mathcal{W}_N$	$2N$	$2N - 1$	$2N - 2$	M

Remark 5.4 (Depth 2 from higher Gaudin). For $k_{\max} \geq 3$, the collision residue encodes “higher Gaudin Hamiltonians”: the standard Gaudin Hamiltonians (1.1) correspond to the $k = 1$ (simple-pole) truncation. The higher poles produce operators that commute with the Gaudin Hamiltonians and with each other, extending the integrable system from the Gaudin model to a larger commutative algebra. For $\mathcal{W}_N$, the integrable system has $\lfloor N/2 \rfloor + 1$ independent generators (one per even-order Casimir of $\mathfrak{sl}_N$ plus the energy-momentum contribution), each of different differential-operator order.

6 The seven-face identification at genus zero

The Gaudin identification is one of the seven faces of the collision residue. The full seven-face chain at genus zero is:

- (F1) *Twisting morphism*: $r(z)$ is the genus-zero binary projection of $\pi_{\mathcal{A}}: B(\mathcal{A}) \rightarrow \mathcal{A}$.
- (F2) *Line-operator R -twist*: $r(z)$ governs the meromorphic tensor product (Dimofte–Niu–Py [3]).
- (F3) *Classical PVA r -matrix*: $r(z)$ is the classical r -matrix of $\text{PVA}(\mathcal{A})$ (Khan–Zeng [4]).
- (F4) *GZ commuting differentials*: $r(z)$ generates the commuting Hamiltonians $\{H_i\}$ (Gaiotto–Zeng [2]).
- (F5) *Yangian R -matrix*: $r(z)$ is the classical limit of $R(z; \hbar)$ with $\hbar = 1/(k + h^\vee)$ (Drinfeld [5]).
- (F6) *Sklyanin bracket*: $r(z)$ generates the Poisson bracket (Semenov-Tian-Shansky [6]).
- (F7) *Gaudin generator*: $r(z)$ generates the Gaudin Hamiltonians (Feigin–Frenkel–Reshetikhin [1]).

Theorem 3.1 establishes $F4 \Leftrightarrow F7$ for affine Kac–Moody. The remaining links are:

Proposition 6.1 (The chain at genus zero). *For affine Kac–Moody at non-critical level:*

- (i) $F1 \Leftrightarrow F7$: the twisting morphism at genus zero is $r(z) = \Omega/((k + h^\vee)z)$; the Gaudin Hamiltonian is its multi-point extension.
- (ii) $F5 \Leftrightarrow F6$: the Yangian quantizes the Sklyanin bracket (Drinfeld 1985, Semenov-Tian-Shansky 1983).
- (iii) $F4 \Leftrightarrow F7$: Theorem 3.1.
- (iv) $F1 \Leftrightarrow F3$: the PVA λ -bracket at leading order is the collision residue (the three-parameter identification).

(v) $F_1 \Leftrightarrow F_2$: the bar-cobar R -matrix is the line-operator R -twist (DNP identification).

The chain $F_1-F_2-F_3-F_4-F_7-F_5-F_6$ closes without obstruction. Each link is an independent theorem; the closure is the coherence of the seven-face programme.

6.1 Elliptic Gaudin via Felder R -matrix and KZB

At genus one, the collision-residue programme upgrades from KZ to the Knizhnik–Zamolodchikov–Bernard (KZB) connection, and the simple-pole kernel $1/(z_i - z_j)$ is replaced by the dynamical propagator $\theta'_1(z_i - z_j, \tau)/\theta_1(z_i - z_j, \tau)$ (FM30 warning: *not* the Weierstrass zeta ζ_τ , which breaks CYBE). The elliptic Gaudin Hamiltonians carry an extra dependence on the dynamical parameter $\lambda \in \mathfrak{h}^*$.

Theorem 6.2 (Elliptic Gaudin from chiral collision at $g = 1$). *Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ and $\widehat{\mathfrak{sl}}_2$ at non-critical level k . The genus-one collision residue of $\Theta_{\widehat{\mathfrak{sl}}_2}^{\times n}$ on $E_\tau^{\times n}/\Sigma_n$ produces the Felder dynamical R -matrix*

$$R_{\text{Felder}}(z, \lambda, \tau) = \text{id} + \hbar \cdot \frac{\theta'_1(z, \tau)}{\theta_1(z, \tau)} \cdot \Omega(\lambda) + O(\hbar^2),$$

where $\Omega(\lambda) = \Omega_{\widehat{\mathfrak{sl}}_2} + \lambda \cdot h \otimes h$ is the dynamical Casimir, and the elliptic Gaudin Hamiltonians are

$$H_i^{\text{ell}}(\tau, \lambda) = \sum_{j \neq i} \frac{\theta'_1(z_i - z_j, \tau)}{\theta_1(z_i - z_j, \tau)} \Omega_{ij}(\lambda) + \text{KZB terms}.$$

Flatness of the KZB connection $\nabla_{\text{KZB}} = d - \sum_{i < j} r^{\text{ell}}(z_i - z_j, \tau, \lambda) dz_{ij} - \partial_\tau(\cdot) d\tau$ is equivalent to the dynamical CYBE for the Felder R -matrix, which in turn is the $g = 1$ MC equation on the elliptic collision residue.

Remark 6.3 (Heat-equation prefactor). The KZB connection contains a heat-equation term ∂_τ coupled to the pairing $\frac{1}{2}(J \otimes J)$. The prefactor is $1/(4\pi i)$ on the diagonal part (matrix entries $a = b$) and $1/(2\pi i)$ off-diagonal ($a \neq b$); the factor of 2 comes from the symmetric-matrix chain rule (FM34a). Heat equation: $\partial_\tau \wp_1 = (1/(4\pi i)) \partial_w(\wp + \wp^2)$ at machine precision in the numerical verification.

Remark 6.4 (Szegő three-term as universal modular identity). The Fay trisecant identity, originally for $g \geq 2$ theta functions, specializes at $g = 0$ and $g = 1$ to the Szegő three-term identity (Fay 1973 Cor. 2.5), which holds universally at all $g \geq 0$. The three-term identity is the modular compatibility that replaces the pentagon in the elliptic and higher genus monodromy computations; at $g = 1$ it controls the Felder dynamical associator, at $g = 2$ it underlies the doubly-dynamical DDYBE. The Szegő/Fay identity is *distinct* from the theta-nulls quartic identity on the theta-divisor (which is a $g = 2$ -specific statement).

Remark 6.5 (Higher genus). At genus $g \geq 1$, the collision residue receives corrections from the higher-genus projections of $\Theta_{\mathcal{A}}$. The Gaudin Hamiltonians generalize to commuting operators on $\overline{\mathcal{M}}_{g,n}$, and the connection ∇^{GZ} acquires curvature proportional to $\kappa \cdot \omega_g$ (the Hodge class). The genus-zero seven-face identification extends to higher genus on the uniform-weight lane; for multi-weight algebras at $g \geq 2$, the cross-channel correction $\partial F_g^{\text{cross}} \neq 0$ introduces additional terms.

7 Compute verification

Computation 7.1 (Engine catalogue). The results are verified by two compute engine suites:

- (i) *Gaudin-collision engine* (105 tests): verifies the identification $H_i^{GZ} = H_i^{\text{Gaudin}} / (k + h^\vee)$ for $\mathfrak{sl}_2$, $\mathfrak{sl}_3$, $\mathfrak{sl}_4$, $\mathfrak{so}_5$, and G_2 at multiple levels; verifies the operator-order trichotomy for all standard families; verifies the Virasoro collision residue pole structure; checks commutativity $[H_i, H_j] = 0$ numerically for $n = 3, 4, 5$ points.
- (ii) *Seven-face cross-check engine* (72 tests): verifies the seven-face chain at genus zero; checks the three-parameter identification $\hbar_{KZ} = \hbar_{\text{DNP}} = \hbar_{\text{bar}} = 1/(k + h^\vee)$; verifies the $\mathcal{W}_3$ collision depth $k_{\text{max}} = 5$ and operator order 4; cross-checks against the Yangian R -matrix and Sklyanin bracket computations.

Total: 177 tests, all passing.

Computation 7.2 ($\mathcal{W}_3$ collision residue). The $\mathcal{W}_3$ collision residue is computed explicitly. The W -self-OPE at the z^{-6} pole is

$$W(z)W(w) \sim \frac{c/3}{(z-w)^6} + \dots$$

After d log absorption: $r_{WW}(z) = (c/3)/z^5 + \dots$ The full collision residue, including the TT , TW , and WW sectors, has the structure

$$r_{\mathcal{W}_3}(z) = \frac{c/3}{z^5} + \frac{\alpha_3}{z^3} + \frac{\alpha_1}{z}$$

where α_3, α_1 are elements of $\mathcal{W}_3^! \otimes \mathcal{W}_3^!$ depending on c and the $\mathcal{W}_3$ structure constants. The compute engine verifies this at $c = 2$ (the free-field realization).

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